

A Deadline-Aware Two-Level Architecture for CAN Masquerade Detection

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Local warning first. Cloud confirmation only when requested.

V2C-Sentinel separates two operational decisions: an immediate level-1 warning from a local detector, and a level-2 confirmation returned through a replayed cellular-delay channel. A token bucket limits cloud requests and exposes the trade-off between confirmation speed and communication demand.



5 ms local warning

29–31 ms best confirmation

150 / 300 ms deadlines

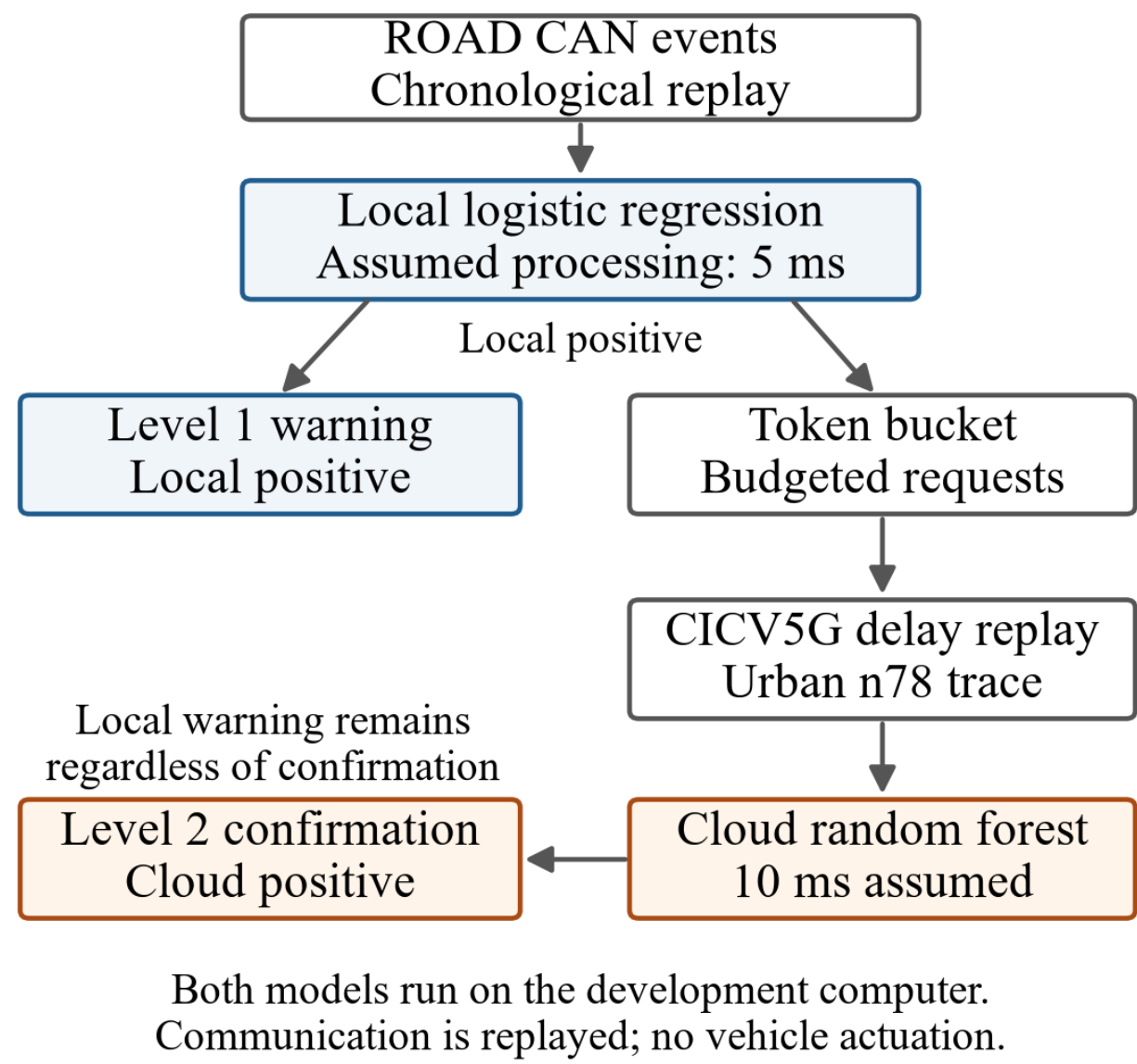
PROBLEM AND CONTRIBUTION

CAN intrusion detection must distinguish malicious activity from legitimate vehicle traffic while producing a warning early enough to support action. A remote model can add evidence, but request admission and cellular delay determine when that evidence becomes available.

CONTRIBUTION

- Preserves every local positive as a level-1 warning
- Sends eligible positives to a separate cloud detector
- Replays measured 5G delay under finite request budgets
- Reports warning and confirmation timing separately

SYSTEM ARCHITECTURE



DATA AND MODELS

ROAD CAN recordings

- Development: 63,260 attack messages; 874,018 normal messages
- Test: 38,003 attack messages; 106,939 normal messages
- One evaluated attack episode and one normal recording per split

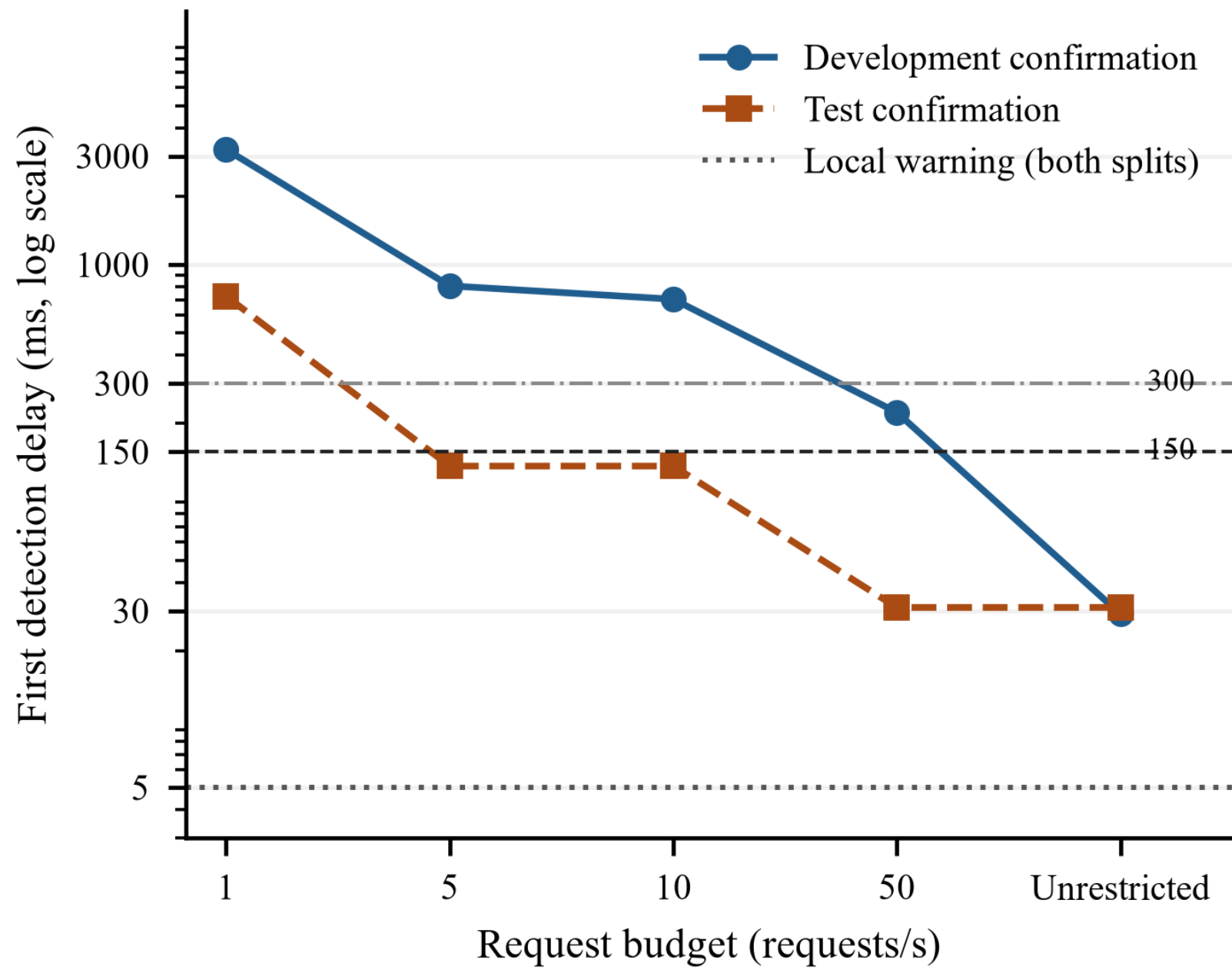
LOCAL Logistic regression; 95th-percentile attack-class threshold; assumed 5 ms processing.

CLOUD Random forest; requests admitted by token bucket; assumed 10 ms processing plus replayed cellular delay.

EXPERIMENTAL DESIGN

Chronological replay with request budgets of 1, 5, 10, 50 requests/s and an unrestricted condition. Each configuration is evaluated at 150 ms and 300 ms reporting deadlines. Local warnings remain visible regardless of cloud outcome.

CONFIRMATION LATENCY



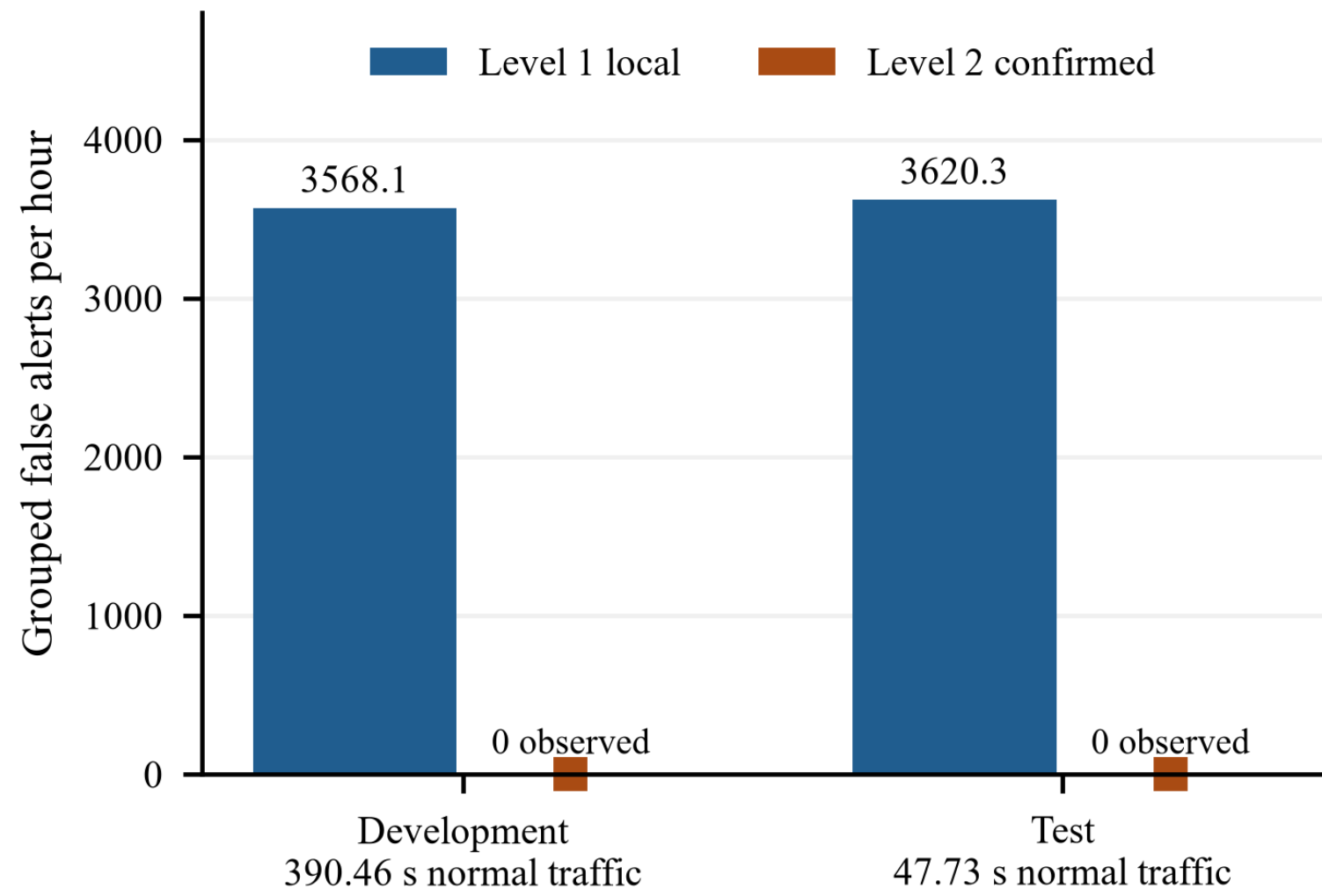
Development confirmation improves from 3.208 s at 1 request/s to 223.6 ms at 50 requests/s and 29.0 ms without admission limits. Test confirmation reaches 129.8 ms at 5–10 requests/s and 31.0 ms at 50 requests/s or unrestricted.

DEADLINE OUTCOMES

Budget	Dev 150	Dev 300	Test 150	Test 300
1 req/s	0/1	0/1	0/1	0/1
5 req/s	0/1	0/1	1/1	1/1
10 req/s	0/1	0/1	1/1	1/1
50 req/s	0/1	1/1	1/1	1/1
Unrestricted	1/1	1/1	1/1	1/1

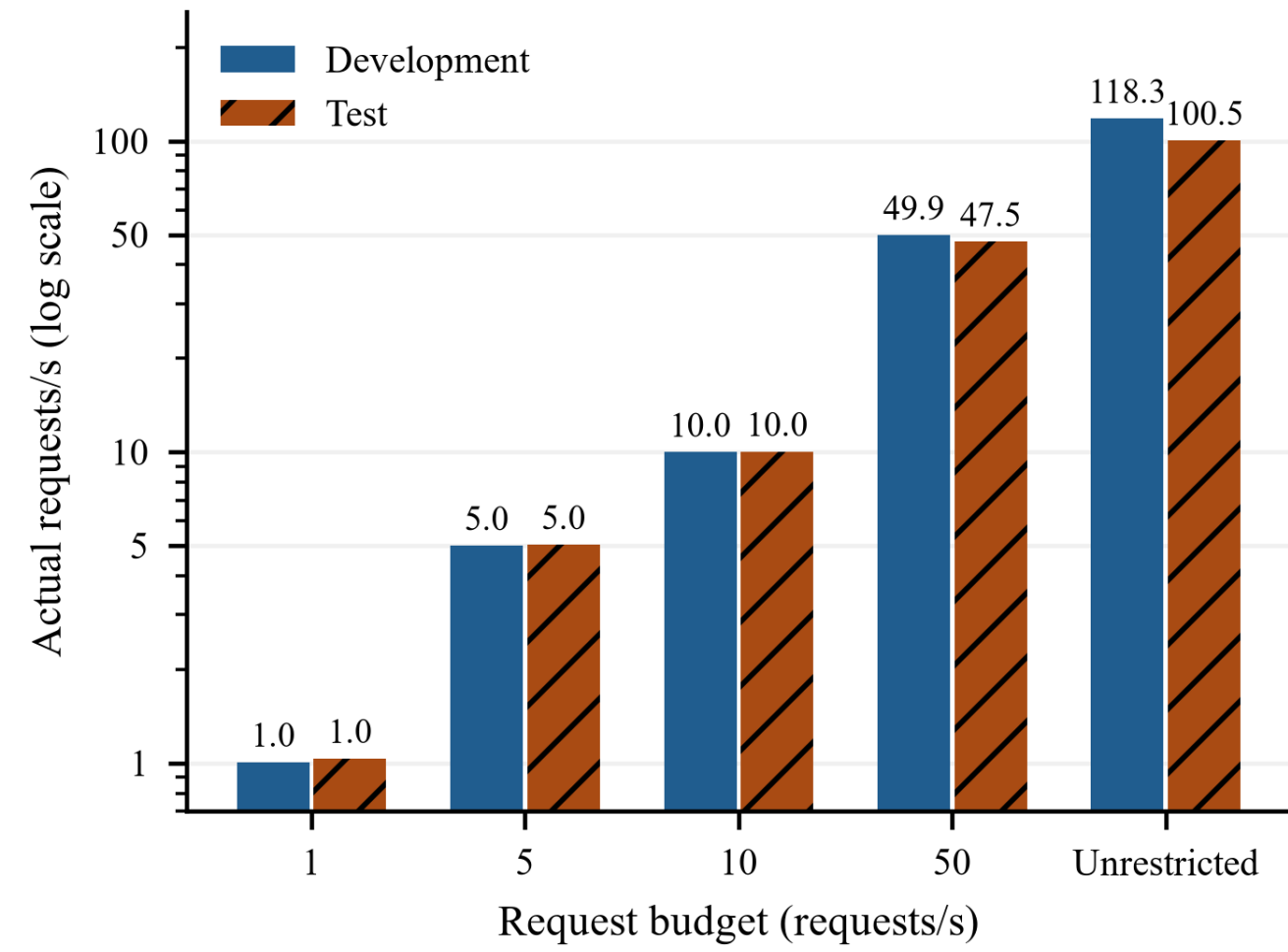
Cell values are timely confirmations / evaluated attack episodes. With one episode per split, these are observed case outcomes rather than population detection rates.

FALSE-ALERT BURDEN



Level 1 produced approximately 3,568 and 3,620 grouped false alerts per hour on the development and test normal recordings. Level 2 produced no observed false alerts on those two short recordings. This does not establish a zero population false-alert rate.

COMMUNICATION DEMAND



Unrestricted replay admitted 49,539 development requests and 6,503 test requests. Finite budgets reduce network load but can delay the first useful confirmation beyond the application deadline.

WHAT THE EVIDENCE SUPPORTS

- SUPPORTED**
- Immediate local warnings in both evaluated attack episodes
 - Cloud confirmation reduced observed false alerts on two normal recordings
 - Confirmation latency depends on request budget and recording

LIMITATIONS

One attack episode per split; short normal exposure; assumed processing times; both models executed on the development computer; no target-vehicle or production-cloud validation.